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Jerry Lettvin is Professor of Electrical Engineering and Biomedical Engineering, Emeritus, at MIT, Cambridge, Massachusetts. Professor Lettvin is well known for his classic 1959 paper with Humberto Maturana, Warren McCulloch, and Walter Pitts, "What the Frog's Eye Tells the Frog's Brain," Proceedings of the Institute of Radic Engineers 47: 1940–1951 [reprinted in Neurocomputing 2]. Some of the history and implications of this influential paper are discussed in the interview.

June 2, 1994, Cambridge, Massachusetts, at MIT [Some material added in 1997]

ER: I want to find out how you grew up and how it was you happened to get interested in the brain. What's your date of birth and where were you born?

JL: I was born in 1920, February 23, so I am now seventy-four.

ER: Where were you born?

JL: In Chicago, grew up and went to school there.

ER: What did your parents do?

JL: My father was a lawyer. My mother was a piano teacher. I was the oldest of four children, and my youngest brother became a concert pianist. He's now professor of music at Rutgers University. My mother failed with me because she wanted me to be a pianist. I gave my first concert when I was eight and then retired from the field and decided to write poetry. That's what I wanted to do, to become a poet.

When I graduated high school, mother made it clear to me that either I would go into medicine or all funds would stop. She had a whim of iron, so I agreed to study medicine. I eventually entered University of Illinois Medical School in 1939 and graduated early in 1943 because of the war. Then I came to Boston City Hospital to the Harvard Neurological Service under Denny-Brown.

But my interest in nervous systems started in my premed college years at Lewis Institute, a working-man's school. Fifteen dollars per course. There I was exposed to psychology by David Boder. He had worked with Pavlov before becoming the psychologist for the prison system in Mexico. He

bragged that it became the most advanced prison system in the world, with conjugal rights and an educational program.

He was a remarkable teacher who started me thinking about psychology as an interesting field, so I began reading. By the time I got into medical school, I was already committed to study the nervous system if I couldn't escape and become a writer. I was very lucky to have Gerhardt von Bonin as my teacher in brain anatomy. You want me to ramble on?

JA: Absolutely.

JL: I was a first-year student and I told Gerhardt I wanted to be a poet, that I really hated medicine. He read some of my poetry and said, "You're absolutely right; you'd make a better poet than a doctor."

So, comes the final practical examination in anatomy, and Gerhardt approaches the table. There's the cadaver. He looks me in the eye. He says, "You know, you want to be a poet, and I think it's right that you should be a poet. Let me put it this way. I will pick up something out of the cadaver. If you name it, you pass and you go on; that's your fate. If you can't name it, you fail, and then, by God, you can become a poet."

So he picks out something from the abdomen, deep in the abdomen. He holds it up without looking at it and says, "What is this?"

At this point, you know, it's a very weird choice. I decide to take the middle ground. So I say, "Cerebellum."

He looks at me with disgust, drops it back in without looking at it. "You named it. You pass." He was a very civilized guy, wonderfully civilized.

The year before I had gone to medical school, I had spent at the University of Chicago. There I met Walter Pitts, who became and remained my best friend. Walter was an autodidact. He taught himself logic and mathematics, and was able to read a fair number of languages, including Greek and Latin.

He already had a peculiar history. At the age of twelve he was chased into a library by a gang of ruffians, and took refuge there in the back stacks. When the library closed, he didn't leave. He had found Russell and Whitehead's *Principia Mathematica*. He spent the next three days in that library, reading the *Principia*, at the end of which time, he sent a letter to Bertrand Russell, pointing out some problems with the first half of the first volume; he felt they were serious.

A letter returned from Russell, inviting him to come as student to England—a very appreciative letter. That decides him; he's going to be a logician, a mathematician.

He went to University of Chicago, which is where I met him; but he never registered as a student. The two of us met at a Bertrand Russell lecture, and for some reason or another we became fast friends. I knew no science whatsoever, no mathematics at all. I was only interested either in literature, or if I was going to have to go to medical school, medicine. The two of us became inseparable. During that year at the University of Chicago, Walter had

gotten hold of Carnap's new book on logic. This was in 1938. He walks into Carnap's office with his own annotated version of the book, pointing out some flaws. And he gives it to Carnap, talks to him a while, then goes out, but doesn't introduce himself. Carnap spends the next couple of months hunting high and low for that "newsboy who knew logic." In the end, he did find Walter and persuaded the University of Chicago to give him some menial job. Walter had no funds, had separated himself from his family, so that was good.

We stayed fast friends, even when I'd gone to medical school. We met often, would have long conversations together. He was a wonderful teacher. When I was in the third year of medical school, McCulloch came to University of Illinois, and Gerdhardt, who had worked with him once before at Yale, took me over to meet him. I remember, when I was about fourteen, I had read Buckle's history of the Scotch mind in the seventeenth century [Henry Buckle, *On Scotland and the Scotch Intellect*. Chicago: University of Chicago Press, 1970; orig. pub. 1857]. It set up in me a morbid fear of the Scotchmen. So when I first meet McCulloch, who carries himself as a cavalier, a Scotch cavalier, he scared the hell out of me.

Nevertheless, I settled in, and I became absolutely committed to the notion of working in the nervous system. I didn't do any experiments at the time. Walter then came over, and Warren was enchanted with him. Walter was homeless, and I had to escape from my family because the supervision was incredibly intense; it's very hard to say how intense it was. And so early in 1942 Warren invites Walter and me to live with him and his family. Warren and his wife, Rook, were always enormously generous. We settled in, and it was in the evenings then that Walter and Warren got together on "A Logical Calculus of Ideas Immanent in Nervous Activity" [the famous 1943 McCulloch-Pitts paper, reprinted in *Neurocomputing*]." Walter at that time was, if I remember correctly, about eighteen, something like that. Walter had read Leibniz, who had shown that any task which can be described completely and unambiguously in a finite number of words can be done by a logical machine. Leibniz had developed the concept of computers almost three centuries back and had even developed a concept of how to program them.

I didn't realize that at the time. All I knew was that Walter had dredged this idea out of Leibniz, and then he and Warren sat down and asked whether or not you could consider the nervous system such a device. So they hammered out the essay at the end of '42. Now, from fall of '42 I, in effect, dropped out of medical school, that is, formal medical school. I didn't appear at classes. What had happened was that a lot of personnel had been taken away by the war. At that time, the military was short on doctors and nurses, so I became, as it were, intern, resident, and nurse, depending upon what was needed, at the Neuropsychiatric Institute, the basement of which housed Warren. The upper floors were dedicated to psychiatry on one side of the building, neurology and neurosurgery on the other. I would appear for examinations in the various courses, but otherwise didn't go very much to class

because it was sometimes close to twenty-four hours a day that I had to work. Walter very decently used to come over and keep me company when I was exhausted and had to rest.

The hospital gave me residence quarters so that I wouldn't have to go home, and Walter still lived with Warren, but he would come in, and the two of us would talk together. I graduated from medical school to everyone's surprise because I'd only appeared for examinations. But now I was really passionate about neurology. I had written a criticism of some stuff that had been put out at Harvard on the Argyll-Robertson pupil, and gave an alternate explanation for it. I sent this to Harvard, along with my application. I was immediately accepted, to my family's astonishment, in Harvard neurology. So I went there.

In 1943, I was working part time in neurology and part time in medical wards because I had to have general medical experience. My co-intern had a relative, a second uncle or something of the sort, by the name of Norbert Wiener, and said we should go over and visit him sometime. OK, so we go over and visit, and I meet Norbert Wiener for the first time. Wiener had just lost his right-hand man to a skiing accident, and was desolate; he couldn't find anybody else of that quality.

So I said, "I happen to know a young man from Chicago a mathematician, who's extraordinary" and so forth, and I described Walter Pitts.

Wiener said, "There doesn't exist such a person."

So I called McCulloch, and I remember the two of us chipped in together and got Walter a round trip train ticket to Boston.

It was very funny. If you don't mind an image occasionally, I remember Walter walking into Norbert's office, and Norbert says, "Hello," and no introductions, "Come, I want to show you something," and takes Walter next door, where there is a blackboard on two walls, and Wiener says, "I want to show you something; I'm going to prove the ergodic theorem." He starts out, and a short distance down the board, Walter begins some critical questions or asking for clarifications, and by the time they'd finished the first board, with Walter going along making commentary, it's sort of obvious what's going to happen. By the end of the second board it's definite, Walter's going to work with Wiener.

Now this was a wonderful thing. When Walter moves to work with Wiener everything is quite nice, except now I have to go off to the wars. After basic training in January of 1944, I have to report to New York, Bellevue Hospital, for a few month's training in neuropsychiatry. Walter at this time is hired by Kellex Corporation, which is part of the Atomic Energy Project. They're over in the Woolworth Building, and so we move to New York together.

We take a room together in the Village. He goes in the morning to Kellex, I go over to Bellevue, but meanwhile we're having a ball because I keep sending him at Kellex postcards written in German: "Enclosed you'll find the secret documents." They never got to him, you know, and we wondered

what happened to them, whether people were dissecting the periods, or whatever.

Any rate, he also used to have fun there. He tangled with General Groves in a mischievous way. There were two wastebaskets in every room, a red wastebasket and a green one.

The red wastebasket was for sensitive material, anything in it had to be taken down, burned in the presence of two witnesses. So one day when the General is coming down the hall, Walter's sitting there, feet on the desk, cracking peanuts, throwing the shells into the red wastebasket, and the General screams, "What are you doing?"

Walter says, "Look at the shells," and on each shell he had inscribed alpha or beta, the magic Greek letters used in formulas, so the peanuts had to be taken down and burned in the presence of two witnesses.

In any case, the people there at Kellex were not allowed to talk to each other. Our apartment was one of the places to which they would come in the evening so as to be able to talk. If anybody had known this, I'd never have been allowed to go overseas, even though I understood nothing of what they said.

You want anecdotes, so I'm giving you some. At this time, the Atomic Energy effort was being embarrassed by the fact that the draft laws gave no exclusion of scientific personnel. The result was that good technical people were being drained from Kellex. They didn't know how to stop it. The 42nd Street induction station was an important place.

Now I always wondered whether the solution occurred by design or accident. There was a particular psychiatrist, Sam Wortis, whom I accused of arranging it. He refused to deny it, but he also refused to affirm it at the time. Installed as senior psychiatrist in the 42nd Street induction station was A. A. Brill, the fellow who had translated Freud. Now, Brill was a nut. He had the theory that schizophrenia had to do with symbolism, and since mathematicians dealt with symbols, etc.

So once Brill was put in office, and mind you, nobody had instructed him, it was his own meshugas that kept things going, scientists and engineers were being turned down right and left for being preschizophrenic, prepsychotic. It was highly demoralizing to the physicists and mathematicians who took seriously this schmuck's diagnosis. At any rate, the morning that Walter was supposed to appear, I decided, "I'm not going to go." It happened to be my turn, I've got to be there, but I call off sick; and it was fortunate I did, because the story came back to me.

Walter at that time wore a beard. He walks in, Brill looks at him and says, "Young man, why do you wear a beard?" And Walter says, "Old man, why do you?" And you can imagine that from this point it was straight downhill. Walter came out with a diagnosis of frank schizophrenia. Now, this upset him considerably. In some respects, he was fragile, and he tried his best to go back and volunteer. It took several people, including myself, to talk him out of it.

Then I go off to the wars and I'm overseas till the fall of 1946. I come back, I'm sort of demobbed, and Walter and Wiener had decided it was about time that I should learn some science.

So Wiener admits me single-handedly here at MIT as a special student. I hadn't had the calculus yet, and here I am, late on, twenty-six years old, and what the hell, what am I going to do? At any rate, I'm signed up for courses 18.01 and 18.02 [elementary calculus], analysis with Franklin, 8.01 and 8.02 [elementary physics], and that's my first term.

Now, I've got to put it to you, I had second-hand contact through Walter with this stuff, but you don't learn it when you are aged, and twenty-six is advanced age. Anyhow Oliver Selfridge, Hyman Minsky (the future economist), Walter, and I all move in together in one room on Beacon Street. The VA has taken me on as psychiatrist, and that pays a good part of our cost of living. I'm trying to make my way through MIT at the same time as doing part-time psychiatric work at the VA. It's not an easy job.

So I managed to survive the first term, passing everything. And then the second term is 18.03, 8.03, differential geometry, a third course in physics, and something else that I've since forgotten. Halfway through the term, Walter manages to lose one of Wiener's manuscripts, which he was supposed to look at. So I say to Walter, "Listen, I'm not cut out to be a scientist even if I've learned something. Let me take the blame." It's clear that Wiener's going to kick me out if that's the case. So I took the blame and got kicked out, and that was it.

So now I was on my own. One of the things I encountered was a weird prejudice. I had graduated from Harvard, internship in neurology; and that was the field I was really interested in. All of a sudden, my New York training and army experience had made me into a psychiatrist, and I went overseas, and became the head of neuropsychiatry at the 237th General Hospital.

I don't regret any of it. It was the busiest goddamn time I ever had, that oversea stint. But once I got out of the army, I come to visit Denny-Brown. Denny-Brown almost doesn't want to talk to me.

I ask, "What's wrong?"

He says, "You became a psychiatrist."

I said, "I didn't become a psychiatrist. You know I'm a neurologist. They forced me."

He said that was no excuse. Looking back, I agree with him. I wanted to go back into neurology, but one thing was true in medicine at that time. Once you were classed as a psychiatrist, forget it; you're never going to be able to get out of it. That classification makes you untrustworthy for anything else. I'm not joking. That was true back in those days. I didn't want to do psychiatry although I had some flattering offers. So I decided to hell with it, and went back to see McCulloch.

He says "Look, the easiest way to get out of this bind is to tell people that you're a physiologist, and see who bites."

I say, "I'm not a physiologist."

He says, "You'll become one."

He was a sweetheart. So I hired out and went to the University of Rochester, and Walter visits me often and we get back together again. At Rochester, they put me in charge of a very strange project, which is "How do you abort cats by using an elevator?" It turns out, a very little known fact today, almost all of the high rides in the amusement parks in the United States were used by women who wanted to have abortions. You see, motion sickness is a well known way of producing abortions. One of the reasons women used to sign up for cruises in the winter is that in the rough seas you get seasick so you abort.

Here's the psychologist who at Rochester says we've got to find out if this works for other animals. So he builds an elevator that moves up and down in square waves, since waves, triangular waves. Anyhow I'm hired on for that. Warren doubles up with laughter, says, "It serves you right."

At any rate, there were also some extraordinarily good people in the neurophysiology of the ear at Rochester. And when the project director is not looking, I'm over there and I'm doing work on the vestibular system of cats, doing, I think, a damn good job. I had discovered some things already, but the boss forbade me to publish it because the data were electrical records. He had no confidence in electrical records, would not allow anyone in the department to publish such data.

So at the end of the year, I got an offer to go to Utah. Gee, that's nice. Meanwhile, I had got married just as I came to Rochester. I had met Maggie in Chicago while I was visiting Warren, and three dates later decided to be married. So Maggie's pregnant now, coming to term, and we're passing through Chicago, and I send a telegram to Utah, "Maggie's having a baby," and I get a telegram back. "We were trying to get in touch with you. Don't come because the funds didn't come through."

Let me describe Maggie for you. She was one of the most beautiful women I ever met, in both appearance and character, utterly unpretentious and with great native intelligence. My family looked down on her, my friends did not. She had had only a high school education, and my mother stayed angry for years.

One evening, shortly after we came to MIT, we visited Giorgio de Santillana, the historian of ideas and an old friend from my student days, at MIT. Giorgio was an adept at interpreting the Tarot. Scarcely a month would go by but Wiener would insist on having his Tarot told. Giorgio vainly explained that the Tarot should be consulted only at times of important choice. Wiener claimed he always had such a crisis and needed counsel.

At any rate Giorgio was charmed by Maggie and offered to read the Tarot for her. She was now about twenty-five. He read the cards with a faint air of disbelief. They told that by age forty she would become a figure of renown, an author and an innovator. Maggie still remembers that evening with some awe, for all came true. In her early thirties, after we had forged

our three kids she was back-ended by a hit-and-run driver. For months she could scarcely lift her arms. Refusing surgery, she studied *Gray's Anatomy* and worked out what sort of mechanical regimen would restore her. Recovery was slow but steady, and within a year she was symptom-free. Others came to her for their mechanical disabilities and she worked out from here newly gained knowledge of anatomy and kinematics some remarkably successful conservative treatment. She charged nothing, was only interested in helping.

Several students, after being helped, persuaded her to hold fitness classes at MIT. Within a year there were about two hundred people per day taking those classes. Then Channel 4 in Boston picked her up, then PBS. For the next seventeen years her program, *Maggie and the Beautiful Machine* (everybody's body), was a PBS standby and her MIT classes stayed crowded. She published four books in her forty's and gloried in the fact that medical practitioners approved of her approach. One book is still in print after twenty-five years. Now she is beginning a new career on the Web, giving counsel on how to relieve back pain without medicaments.

The only pity is that Giorgio could not know that all this happened. He began failing before her ascent picked up steam.

We are still in love, though I am seventy-four and she is sixty-seven. She has not ever changed in my eyes and I cannot, in retrospect, imagine a luckier choice. (Nor, she claims, can she; I had all the luck).

So going back, there I am, high and dry in Chicago with Maggie, and I ask myself, "What am I going to do?" I don't want to be a psychiatrist, but what the hell. So I go to Manteno State Hospital, near Chicago. The head of the hospital is a very decent guy. I say, "Look. I'm willing to trade you psychiatric work if you'll give me a place to put up a laboratory."

So he said, "Sure, no problem. We have a shortage of physicians."

So I moved to Manteno State Hospital, where I was for the next three and a half years. Now what happened was that von Neumann, whom I had met via Warren, heard that I was doing this. He was very decent; and arranged that I get five thousand dollars for equipment. By this time I'd learned enough electronics, so I was able to build my own apparatus, and I started building my own amplifiers, remaking old oscilloscopes, and so forth. Within a few months I had a decent lab going. Some of the work I had done in physiology at Rochester had prepared me. McCulloch was a very good friend, so the laboratory got some attention. I was visited by some remarkable characters, like Pat Wall from Chicago, who became a collaborator.

We formed a little enclave down in Manteno, and it was a delight. Walter would come and stay with Maggie and me for weeks at a time. I worked in Manteno for three and a half years. By this time I've developed my own reputation. At RLE [the Research Laboratory of Electronics at MIT], Wiener convinces Jerry Wiesner that it's about time he got some physiologists of the nervous system.

Warren was a distinct catch because he had gotten tired of Chicago. So Warren, Walter, Pat Wall, and I sat together and decided that if they would take us as a quartet, we were willing to go, so we came to MIT. That was in 1951.

By this time I had gotten some sophistication in circuit design. I had to because it was the only way in which I could work, and in fact, I had built some things that had already attracted some comment. I had no problem settling in here and getting started. Walter and I now were more or less inseparable; Maggie was as fond of Walter as I was.

At MIT McCulloch became seduced into what can be done theoretically with nerve networks. A number of people gathered around and tried to see what could be mined out of this.

Walter by this time had more or less set himself against the concept of doing only abstraction. To him it was much more important to come up with notions of how automata were to be devised. He wrote a thesis, a very long one, on the properties of nets connected in three dimensions. Others like Caianiello had already done two-dimensional nets. Walter had come up with some very strange properties of the three-dimensionally connected net. He was in no uncertain sense the genius of our group. He was absolutely incomparable in the scholarship of chemistry, physics, of everything you could talk about history, botany, etc. When you asked him a question, you would get back a whole textbook. You sat back and listened for two or three hours because he would go on and on. To him, the world was connected in a very complex and wonderful fashion.

At the same time he was strongly opposed to having his name known publicly, so much so that when they offered him an advanced degree at MIT if he would just sign his name or translate a page from the German, which he did very easily, he refused. Later on, when they offered him an official position if he would just sign his name to a document, he refused.

In many respects, he was like the eccentrics you read about in England. He had exactly that quality. But he was a most winning person. I mean, almost everyone who knew him was fond of him. He was in a sense almost pure thought, thought personified, but with a delightful understanding about things generally, and was a most amiable companion.

At any rate, there was a sudden violent turning of Wiener against McCulloch due to Mrs. Wiener, who hated Warren, and Wiener cut off all relations with McCulloch and anybody connected with McCulloch, which meant Walter as well. Now Wiener was what Walter never had, a father figure, and that threw Walter into a decline from which there was no pulling out. He burnt the manuscript for his work on three-dimensional nets. Jerry Wiesner offered me a fair amount of support for the lab if I could just recover bits of it, but I couldn't. Walter was quite complete in destroying his own past work.

From that point on, we had no way of getting him interested in things. He did one major piece of work with Bob Gesteland. Bob Gesteland's first work

on olfaction published in the *Journal of Physiology*, a really prestigious work that started a new branch of study, was done in collaboration with Walter. That was it.

When Humberto Maturana and I did the work on the frog vision, Walter was a little appalled by the results, which were very different from what he expected. That is, here were the qualities sought, but not as you would have them. There were invariances that were not formally tractable, or at least not in any ordinary logical way, and that were of a nature different from anything that anyone expected. They looked more like black magic than anything else.

Walter had an interesting response to these results. It's hard to characterize, but let me try to do it. On the one hand, he believed the results whole-heartedly, and he approved vastly, but on the other hand, it was an index to him that logic was not the right approach to the brain. See, up to that time, Walter had the belief that if you could master logic, and really master it, the world in fact would become more and more transparent. In some sense or another logic was literally the key to understanding the world.

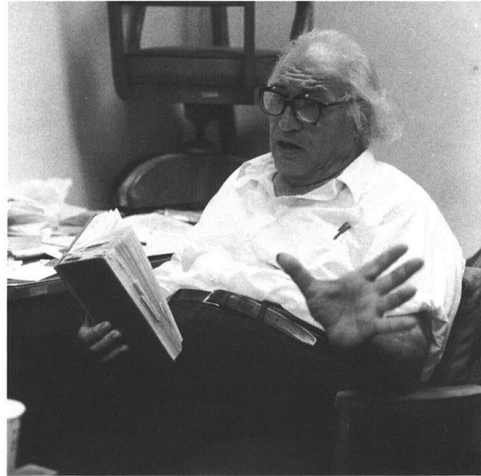
It was apparent to him after we had done the frog's eye that even if logic played a part, it didn't play the important or central part that one would have expected. And so, while he accepted the work enthusiastically, at the same time it disappointed him. He would never admit it, but it seemed to add to his despair at the loss of Wiener's friendship.

Warren, as I say, had committed himself to two-valued and three-valued logic. Walter lost interest in that. He was perfectly willing to be amiable about it but he didn't want to spend too much thought on it. He became more and more introverted. It was difficult to find him. He would try to escape from all his friends. We'd go hunting for him night after night.

Watching him destroy himself was a dreadful experience for several people who knew him very well. Warren, at the same time, also began having trouble. He had developed an inability to eat solid food. He had had two lung episodes, blowing out a hole in the lung and getting a pneumothorax. They stitched him up, and then finally attached the lung to the wall of the chest. But in doing so they set up what appeared to be a vagovagal reflex. Every time he would eat solid food his cardiac T-waves would undergo reversal, indicating coronary insufficiency. He would only take liquid food. It was clear that he felt that he was decaying in his physical powers. To the end, he always was enthusiastic, but you felt that somehow or another you weren't touching him. So Pat and I and Humberto worked more or less independently.

ER: Could you discuss the beginnings of artificial intelligence?

JL: Let me now say that I know nothing about artificial intelligence, which is a separate discipline from nervous physiology. I have admired the effort since Marvin Minsky and Seymour Papert began it at MIT, but I never tried to study it and have as little grasp of it as anyone not directly involved. I do



know a little about the history of the problem that they propose to solve and I can talk about that.

When Walter and I went to live with Warren and Rook McCulloch in 1942, Walter introduced Warren to Leibniz's remarkable work in the latter part of the seventeenth century. First of all, Leibniz showed long before Boole that logic reduced to arithmetic. The demonstration was incomplete but went far enough to establish the plausibility of the hypothesis.

Leibniz was an engineer as well as mathematician, scientist, and philosopher. Early in his career he designed and built the first calculator that did all four arithmetical operations in decimal and thus superseded Pascal's adding machine that could only add and subtract numbers. Leibniz's method of multiplication and division was ingenious and remained in use for hand-powered calculators till the end of the nineteenth century.

But then he invented the binary number system whereby multiplication and division reduced to addition and subtraction. He was unable to build a hand-powered calculator in binary because of the friction between the many parts, and the technology of the time could not support alternative designs (e.g., there were no reliable valves).

Nevertheless he pointed out that if a binary calculator was possible, so was a logical machine which could perform any finite task that could be

expressed completely and unambiguously in logical language. That was the proposition that Warren and Walter adopted. If a neuron could be axiomatized as a device that performed elementary logical operations, then a nervous system would be regarded as a computer.

The thrust of their original paper on "The Logical Calculus of the Ideas Immanent in Nervous Activity" (and of their elegant but less often quoted second paper, "How We Know Universals") [both in *Neurocomputing*] was to apply this concept.

Half a decade earlier Turing had issued his fundamental paper on the universal logical engine. Half a decade later, at the end of WWII, von Neumann and Bigelow, would begin to design the first electronic computer in this country. So in that magical decade the great effort began to realize Leibniz's concept.

Now Leibniz had also developed the ideas of negative feedback control. The practice of this control goes back to antiquity, but Leibniz abstracted the idea, and expressed it in intelligible form. Wiener's "Cybernetics," appearing in mid-decade was, as he says in the book, also in the tradition of Leibniz. And so too was the concept of information that had appeared at the hands of Szilard in the 1920s and was to be elaborated wonderfully by Shannon and Weaver and other mathematicians beginning in the '40s.

What with information theory as analyzed and logical machines as synthesized it was certain that Leibniz's logical automata would appear, machines that performed tasks and acted as if animated by intelligent purpose.

While I could not but be aware of these potential developments by close association with Warren and Walter, it wasn't till about five years later, after we had come to MIT, that I began to suspect that logical machines may not provide useful models for perception. Von Neumann had voiced his doubts in his Hixon Symposium Lecture (1949). Certainly I had not the competence in logic or mathematics to consider the issue myself. But it seemed to me that before you considered a mind that would perform combinatorics on perceptions there had to be some qualifications on how perceptions represented the world. Everything I had found so far and had read about suggested that the world was reported by natural language rather than logical language, that is, in terms of things and their relations rather than sense data to be processed into the concepts of things. There was no feedback from brain to retina, so that whatever synthetic a priori [genetic programming] was involved in vision had to be indwelling in the retina itself, a five-layer device alternately connected vertically and horizontally in the layers. That a fixed five-layer system of elements, laterally interacting, gave the content of perception to a frog suggested that we were dealing with a system of active complex analogue filters rather than a system of logical elements. Each cell in the array was a complex processor of local spatiotemporal relations in the images, and was modified by remoter influences.

For this reason I had my doubts about artificial intelligence based on logical processing. I realize, of course, that any continuous function can be

represented by a logical program, but also that combinations of such functions, themselves modulating one another, are not so easily represented in real time. But then, my ignorance of computational arts is unsurpassed at MIT and I doubt if I could word a satisfactory argument.

Having delivered this apology let me say that artificial intelligence as a way of understanding and realizing purposive action and even reasoning is entirely in the Leibnizian tradition. It is a noble effort to develop by synthesis new principles to guide the analysis of animal and human activity. Furthermore, it is in the spirit of biology to discover functional laws as well as composition of parts and the programs encoded in DNA. In short, AI becomes an adjunct to psychology in the normal course of events.

I would like to put it that AI and computer science are one and the same were it not that this would incur the wrath of my colleagues who see a great difference between their science and psychology.

But the changes in thought brought about by the scientific revolution in the seventeenth century have not only persisted but been sharpened. Everything observed must be explained in terms of mechanism alone, everything but the knowing observer. What we call process, the strategy of performing a task, has to be explained by the state-history of the mechanism, that expresses the process. In the modern biological view it is a vulgar error to imagine Big Blue as playing chess with the purpose of winning a game. The state-history of the mechanism, which includes the changes made by the opponent's moves, provides a determined sequence that explains how the end-state of the board resulted from the initial state. Nothing is added to our knowledge by the concept that Big Blue is playing chess.

This may sound like a caricature, but any review of modern biology shows preoccupation with mechanism even in the study of brain and nervous function. Whatever studies of process there are, they appear as speculations in clinical journals. And you only have to read the Harvard version of Darwinian evolution to see the lengths to which academics go to avoid the concept of an evolutionary process; or else read the current neurobiology literature to marvel at the utter lack of interest in process.

Ever since biology became a science at the hands of biochemists it has carefully avoided or renounced the concept of purpose as having any role in the systems observed. Purpose in observed nature was once and for all anathematized by science in the seventeenth century. Only the observer may have purpose, but nothing observed is to be explained by it. This materialist article of faith has forced any study of process out of science and into the hands of engineers to whom purpose and process are the fundamental concepts in designing and understanding and optimizing machines.

Leibniz showed beautifully at the end of the seventeenth century that in any system provided with a flow of energy there were two aspects of energy. One explained interaction in mechanical dynamics. The other explained how an interaction would be represented by an independent "spontaneous" flow. E.g., if you cut the cord suspending a weight (in his example), the weight

falls of its own accord; the cutting only removes the impediment. Cutting does not act on this weight, the weight does not act on the cutter. The only relation between the cut and the fall is that the fall represents the cut. The same is true for a flow of water from an elevated tank through a valved outlet. The flow represents the position of the valve. Thus switches and analogue amplifiers are responsible for process.

Once representation, as the essence of spirit, is given an energetic explanation as information, the structure of an automaton as the mechanism sustaining the processing of information becomes intelligible. Logical engines, computers, are possible, and, if possible, cannot be excluded from reality. So far as Leibniz was concerned, if automata would be manufactured, animate things could be regarded as automata.

An automaton, thus, must have two explanations, one in terms of mechanism and accounted by efficient cause, the other in terms of process and accounted by final cause. Thus, to deny purpose to living creatures is foolish, it is tantamount to denying what you don't choose to explain, a peculiar stance for a scientist. Logical computers can be built by the arrangement of switches. Analogue processors can be built by the arrangement of amplifiers. What is the problem? Every process must have a mechanism to sustain it. And this holds even for thinking about a process. The spirit has no subtle action (such as thinking abstractly) which is not accompanied by the action of devices in the brain which is part of the body.

All this is part of a longer essay I will write sometime, organizing all the relevant material out of Leibniz's essays. He felt that living automata differed from manufactured automata in that the elements in living automata were themselves machines while those in manufactured automata were not. Thus analogue computations were possible to living machines; manufactured machines did logical computations.

In the modern case artificial intelligencers took the strong engineering position. Tell us, they say, in unambiguous terms the tasks you want performed, including the task of designing task-performing programs, and we will design them. Even better, we will design them to optimize themselves to the purpose, to learn. We will even build a program that will discover theorems of which it hasn't been informed. That was the salutary beginning. But the difficulties were soon apparent. For example, the world as it appears to us is not unambiguously described anywhere. Idealizations that are natural to logic are unnatural to me as an animal. Every attribute of a definite object of perception, such as color, shape, size, form, order, whatever, is known by relations with other things and changes as they change. Attributes are not as we speak of them, predicates of things in themselves, they are relations. Furthermore, we perceive things in terms of the uses we make of them rather than as what they are intrinsically as things alone.

Yet the current doldrums of AI are far more laudable than the cop-out of biology's anathema on purposive process and its self refinement. It is obvious by now that the immediate future of biology is in mechanism only.

It will classify programs and then explain subassemblies of mechanism as shorn of purpose. Strategy and process will remain, as always, the forte of engineers who are not at all self-conscious about not being scientists.

You see, to me, biology is a dead end now. Current neurobiology is like somebody coming in with shelves and shelves of transistor manuals and saying, "See, this is how, this is what the world reduces to." And you say, "But tell me how to put them together."

At this point, I part company with biologists. I find the artificial intelligencers, the nerve-net people, and their colleagues far more clever. It doesn't matter that they are wrong in some sense; they are right in spirit, and whatever they discover, it's going to be useful in the end, whereas so far as I'm concerned, whatever the biologists are discovering tells you nothing about the nature of the system. I mean, the components don't tell you anything about what the system does, only how it works.

So, from my point of view, I'm a strong supporter even if I'm not a sympathizer or participant in artificial intelligence. Forget about biological preoccupation with receptors and transmitters and magic molecules. You take that for granted as mechanism, but the process of a system is a different thing from its mechanism.

As I say, Walter held this view, but at the same time, by holding that view he partly sterilized himself. The economy that we have encountered in nervous system operations drives us up a wall. Most of that stuff I haven't published, and I'm not likely to publish. But the economy is so fantastic that at times you almost sort of believe Leibniz's notion about the infinity of operations. Somewhere in between the logical approach of artificial intelligence and the semilogical approach of nerve nets lies the possibility of VLSI of analog devices with variable connections. Let me give you an example.

Consider the eye. You have a very interesting item from Pascal's memoirs in the seventeenth century. He buys a telescope. This is only a few years after Galileo introduces it. He's looking up at Venus, and sees it has a phase, like the moon, just as Galileo remarked. He calls over his mother, who is a very intelligent woman. She knows a great deal about science. She looks through the telescope and says, "But Venus is the wrong side to. It's reversed."

Pascal is startled, and says, "How do you know that?" She says, "Because I can also see it with my naked eye."

Now, there's a very interesting point here because people had claimed to see things by naked eye that ordinarily you would say it was impossible to see. You certainly have the identification of the largest moon of Jupiter, Io, by a Chinese astronomer in the third century A.D. There were people whose ability to resolve double stars was quite fantastic. So you look at Pascal's story, and you're not willing to dismiss it for two reasons. First, the *Scientific American* didn't exist then telling you about the phases of Venus. Second, you know that telescopes invert. That doesn't come to mind usually when you're looking up at the stars.

So the question arises, would it be possible for Madame Pascal to see what she did? It's an interesting question. Now, the angular size of Venus at its closest approach is a little over the diameter of a foveal core. It's bright enough that you can use your fovea. At its farthest distance, it's a little less than a cone in diameter. So how are you going to issue a statement that it's flattened on one side?

I looked at this, and then I said, "Well, suppose we imagine the initial layers of the retina as a Wiener filter" [the rods and cones along with the horizontal cells]. Let me specify what I mean.

For a normal young person with a four-millimeter aperture pupil [the optimum aperture for the eye as an optical instrument], a point spot of light at infinity goes into a Gaussian distribution on the retina of about four cones, five cones in diameter, maybe a little more. This is a blur. Furthermore, when you consider the rods and cones, you find that they are resistively connected at their bases to each other in such a way that the space constant of the resistance away from a photoreceptor is even larger than the space constant of the Gaussian distribution.

So you say, God is playing games with us. Here's this woman sees this thing, and you're giving me one form of crapping it out by the optical image and a second form of crapping it out in the receptor arrangement and this is ridiculous.

But there's an interesting point. If the resistive connections are between the cones and the cones are taken as voltage sources, then the current flow into each cone is the Laplacian of the voltages. The Wiener filter consists of a Laplacian applied to a Gaussian, you get super-resolution out of it. With this super-resolution, you now have a sharpening of the image in a satisfactory way. Helmholtz said about the human eye, "If somebody brought me this as an optical instrument, he would be fired on the spot." To put it bluntly, it doesn't even have the optics of a Brownie camera. So far as Helmholtz was concerned, this was very bad optics, but he did not know of image-processing in this way. If you're going to do resolution on the basis of a Wiener filter, wait a minute, that's a better way to do it because in fact I have a discrete manifold of the cones, I don't want to limit myself by discontinuity of receptors in determining the resolution of the image.

I want to be able to get honest-to-God sharpness, which means that I must process at the first layers before I ever go through any nonlinear synapsis. Now I haven't published this. I'm looking for a graduate student who's willing to work on it. It's not a very challenging topic, and image processing experts who say, "Oh, come on, this is obvious," But I'd have to reply, "Well, if it's obvious, let's set up a synthetic model of the receptors on the computer, and let's see what we come up with."

You don't want to try for something that's going to restore the image because that's a different problem. You want to know, "Is the information that I get sufficient to be able to say that I can resolve such and such?" That's all

you want to know, for the point is to account for expert vision and how it is possible.

If what I said is true, then the notions that we have of image processing in the eye are going to have to undergo a major change. You would have to stop working with the computers and start building instead the arrays of coupled analog devices that are capable of giving you this sort of operation. Because if Madame Pascal spoke the truth we had better use the process to tell what to look for in the mechanism rather than the other way round.

JA: That's a wonderful example. You have all these receptors electrically stuck together. They're all interacting, and they should make a mess out of everything.

JL: Which is precisely what they do if you follow ordinary visual physiology. The moment you bring in any sort of additional hints of process, it begins to look tasty. But I'm seventy-four years old. You know, I'm not an adept. I've seldom used computers. I'm not going to learn programming. But the problem is straightforward and any reasonable student should be able to sit down and grind this out in a few months. After all it is now accepted that the human eye has super-resolution, and this has to be accounted in the first layers of the retina for it would be impossible to do later.

The problem I mentioned is the sort of thing that you would say is meat for any engineer. The difficulty is, it's so trivial formally that nobody wants to pursue it, and on the other hand, at the level of the physiology it's at odds with all the received wisdom.

JA: This sounds a lot like ideas I've read of yours along the lines of when the nervous system does something, it really does it right. It's just a question of finding how it's doing it right.

JL: And how to do it economically.

JA: That's a very profound way of analyzing the system.

JL: Let me give you another example. Humberto and I described the bug detectors [in "What the Frog's Eye Tells the Frog's Brain," 1959]. What we did not report, and what to me is still the most astonishing thing about the bug detectors, is the following property which I explored with Arthur Grant. You have a central region of a bug receptor three degrees in diameter, a three-degree visual angle outside of which you get inhibition of response.

So working strictly within this three-degree angle, by masking off everything else, you find the following property. You bring one spot in, move it into the field, and as long as you move it around, wonderful response.

You bring in two spots; if they're rigidly coupled in their motion by a fixed distance between them and are moved around, you get a good response almost as if they are only one spot.

You bring three rigidly coupled spots in, and it doesn't matter their size or their disposition or their distances from each other: move them around as a rigidly coupled triad and there's no response at all.

"Wait a minute," you say "Just a second. You're telling me that this thing is able to separate four phases from three phases?" If you have a white background and two black spots, OK; three spots, forget it. You connect any two spots of the three by a barely visible black line, and all of a sudden it's a two-spot system. It becomes visible. Or else you move any one spot with respect to the other two, now there's a response.

How do you build in a system such as the retina, a device that distinguishes between four and three phases? That to me is a delightful problem. I've been worrying about it for a couple of years, not seriously, you know, not devotedly, but how would you go about it?

But I don't have any students, and I cannot take on graduate students. Because of my age, taking on a graduate student is equivalent to asking him or her to take a chance. If I die, the student's left high and dry. Who else is going to take him on?

What I'm saying is that if you're going to look at visual form even in a frog you're also going to have to use notions of topology not geometry.

Those three spots, if they move translationally, rotationally, it doesn't matter how they move, how big they are, how separate they are, whether they are the same size or different sizes: three spots, no response, providing they move rigidly together. Now that, I consider that to be a lovely find, but I can't publish it, you know, so what's the point? I can see an editor saying, "So big deal. So what does this tell us about vision?" It tell us that in the analysis of vision, if we stick to conventional methods and conventional processing, we're not going to get anywhere.

JA: Is there any work on theory that you've particularly liked recently? You mentioned that you talk to Marvin Minsky.

JL: Oh, yeah. Gad Geiger and I are working on something quite different, top-down sort of stuff. Eight years ago we showed that dyslexia is learned, not genetic or neurological. And we also devised visual tests by which you could demonstrate this. Then we went in and showed that we could teach dyslexics to read. The theory is what our colleagues have jibbed at. If we had only presented the results, everybody would have said, "Oh yes, very interesting." But what we have is evidence that there is an internal feedback of intended action onto the perception of the objects to be acted on. It performs a weighting function, not on anything related to contrast or anything else, but on some operation that switches between the distinctness of form and the indistinctness of texture.

You can demonstrate lateral masking in yourself by experiment. On a clean sheet of paper make a small X in the middle as a fixation point. About an inch and a half to the right or left print a capital letter such as N, about a quarter-inch high. When you gaze at the X you have no trouble in seeing and identifying the N. So your visual resolution is sufficient. Now print two more capitals of the same size on each side of the N to make a five-letter word, e.g., TENET or SANTA or FUNKY. Now when you gaze at the X

most of you will not see the N identifiably even though you know what you printed. It is not as if the N is blurred. It has become nonidentifiable—it has texture but not form. You see clearly both the beginning and ending letter of the word, and surprisingly, the end letter is clearer even than the beginning letter. But it is as if the other letters somehow interfere with seeing the middle letter. Since you can recognize the N by itself, the problem is not in resolution. Instead the order has departed from the form, leaving only a statistical impression rather than a distinct set of relations. It is this conversion from form to texture that is controllable, but not by the will directly, rather by choice of task. It is an unusual concept, but we have evidence to support it and can even demonstrate an appropriate mechanism in the variable receptive field sizes of single cells in frog texture.

In any case, the concept suggested a measure for diagnosing dyslexia and a treatment to alleviate it. What we showed is that lateral masking applies to whatever in the field of vision is not directly relevant to the task you want to perform. It is as if when you decide on the objects of the task, those objects take on distinct foreground form, and everything else is reduced to background texture. This is what is meant by attention. The point, of course, is that attention is not confined to the direction of gaze. It can wander over the whole visual field. If you mislearn where to attend for a task, the region that should be attended is laterally masked and so there is a barrier to correcting misattended task performance.

So we've used that, you see, and now our latest paper has just been published in *Vision Research* about the work we did in Tuebingen [G. Geiger, J. Lettvin, and M. Fahle 1994, "Dyslexic Children Learn a New Visual Strategy for Reading: A Controlled Experiment *Vision Research* 34; 1223-33]. We did it there with grammar school kids. There they let us take a batch of grammar school kids who were dyslexic, and within eight months they were reading within three quarters of a grade of where they should have been reading. The method works. But the theory behind the method, namely, this concept of task-determined later-masking operations, goes against the grain of every psychologist. And yet now there is evidence that this occurs with all forms of perception, for example, audition. We've played our game out, and we wrote sort of a manifesto. Now I would like to switch to some other stuff.

ER: Where would you point people just entering the field now, and what do you think the difficult problems are?

JL: Well, to me, the difficult problems are how to go from an image to its content. The important thing that you realize, for example, in perception is that just as in mathematics, lines are imagined, so they are in vision. You argue from phase boundaries; boundaries are the things that are important in vision. A line is a doubly bounded area. Line drawings are sets of doubly bounded areas.

JA: I get the feeling you aren't too impressed by the very empirical nature of neuroscience now.

JL: I think much of it is beside the point if you want to relate mental activity to nervous activity. Don't misunderstand me. I've great respect for you if you say, "I'm going to find out how this thing is or what the elements are in building it." That's a legitimate project, and if people who are doing it confine themselves to that, if those were the only claims they made, I would be delighted and show intense respect. But the pure nonsense that comes out as if, for example, particular feelings, things, ideas, images were functions of particular chemical compounds, you know, I have no use for that.

JA: I get the impression that neuroscience is far more empirical now now than it was twenty-five years ago.

JL: It is. Twenty-five, thirty years ago it was still imaginative on a systems level. Now unless I were interested in chemistry, I don't think I would go into it at all.

JA: You know, when I was a graduate student at MIT, I found your approach to doing science really inspirational. You did stuff that you thought was important and not what other people thought was important.

JL: Yes, it's going barefoot. You see, I'm lucky. I'm one of the few people who managed to be on the outskirts of everything at the right time. So, in a certain sense, wherever I walked barefoot, I would pick up some things. I don't think that sort of liberty is going to be anything that you get anymore, so I was very fortunate in being a hanger-on in the right places at the right time.

JA: I remember they had the faculty come in and give talks to us incoming students. We had a number of very earnest talks about biology. Then you came and made it sound like a lot of fun. You did interesting problems and had a lot of fun doing it. It was wonderful.

JL: Actually, there are a huge number of delicious problems in biology on a macroscopic level that are going begging because, in the first place, NIH [the National Institutes of Health] will not sponsor anything that doesn't sound like careful science, and in the second place, it's very difficult to get people to be interested in naturalist approaches. It's not deep technical stuff that appeals to them. There's this intermediate range in which I operate sort of solo, simply because I'm not learned enough for one group, and am not enough technically sophisticated enough for the other.

JA: There should be more people like that!

JL: There's a big hedonistic element in looking at individual problems and asking, just as a first-order pass, "What could I do quick and easy before going into it deeply?" These first-order and second-order passes are themselves informative, much more so than people think. The Pascal's mother story, that's a first-order pass at an image-processing system. But it suggests that if you're really going to do image analysis, you could do worse than design yourself a receptor region that is a discrete manifold but in which you use operations of the sort I described, where you're not precise in im-

age down to single points and where you must use something like a Wiener filter. It's a trivial thing to design in analog computers because all you have to do is replace every cone by an op-amp, and you have a voltage governed by the light while the lateral resistor connections between op-amps gives you the current. Now, it's trivial to build such a thing, and behold, if you monitor the currents, you now have a resolution that is very good indeed. It may seem a mess, but it's a sort of mess that tells you what to look for.

So.

